

CPACT: Product Intercept learnings and Compliance measurement for Diagnostics Standardization

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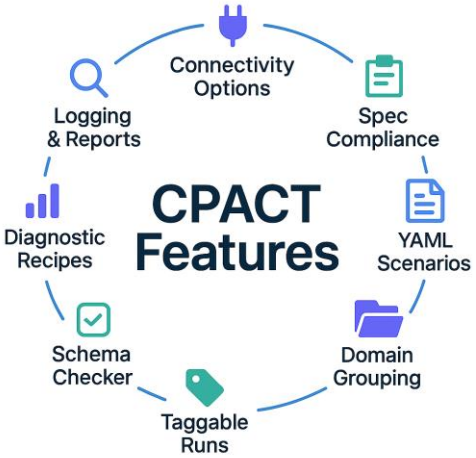
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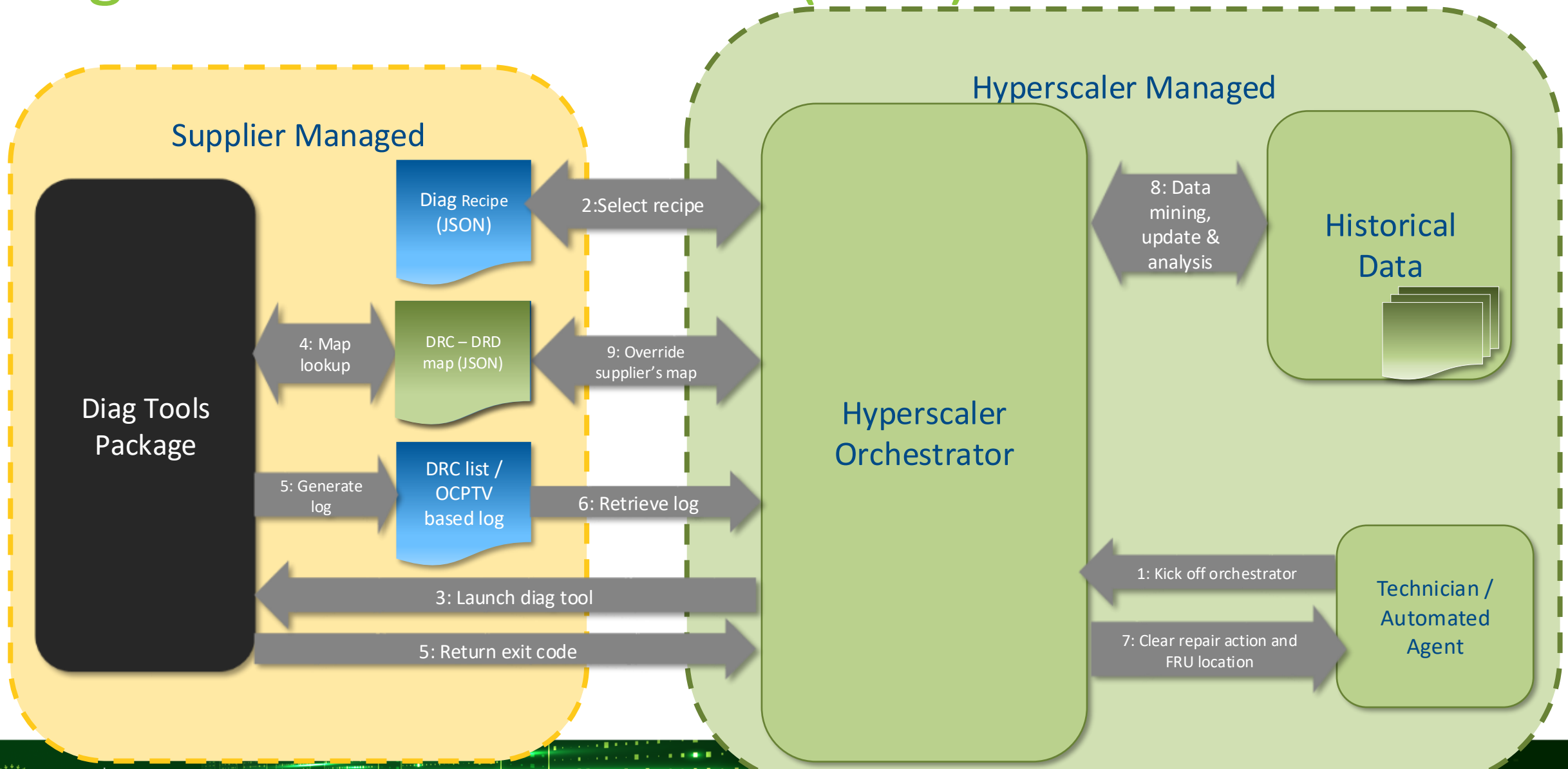
The Compliance Tool Motivation

- **The CPACT compliance tool ensures suppliers adhere to:**
  - Hyperscale CPU Impactless Firmware Update Specification
  - Hyperscale CPU RAS, Debug & Diagnostic Specification
- **Who benefits?**
  - Hyperscalers – Get reports to boost confidence of supplier collaterals
  - CPU Suppliers – Get to build collaterals for all Hyperscalers based on common requirements
- **Design principles**
  - Based on OCPTV guidelines
  - All x86 and non-x86 processor-based platforms
  - Multiple access methods including varied inband & out-of-band support
  - Spec defined schema based – build assistance & execution of compliance scenarios
- **Updates**
  - This year we continued focus on diagnostic specification
    - Progressed from 0.7 to 0.9.





Diagnostic Flow Overview (v0.9)



Diagnostic Interface Enhancements

- Diag Recipe
  - Include recipe classification and other attributes to facilitate recipe selection for Hyperscaler orchestrator
- Diag Result Code (DRC) – Diag Result Detail (DRD) map
  - Incorporates all information related to a DRC in one file
  - Serves as documentation at static time; a dictionary at runtime
  - Provides flexibility for Hyperscalers to apply their own business logic (i.e., repair actions)



Sample Map

- Diag Log (extension to OCPTV output format)
  - Seamlessly incorporate and leverage industry standard log format



Sample Log

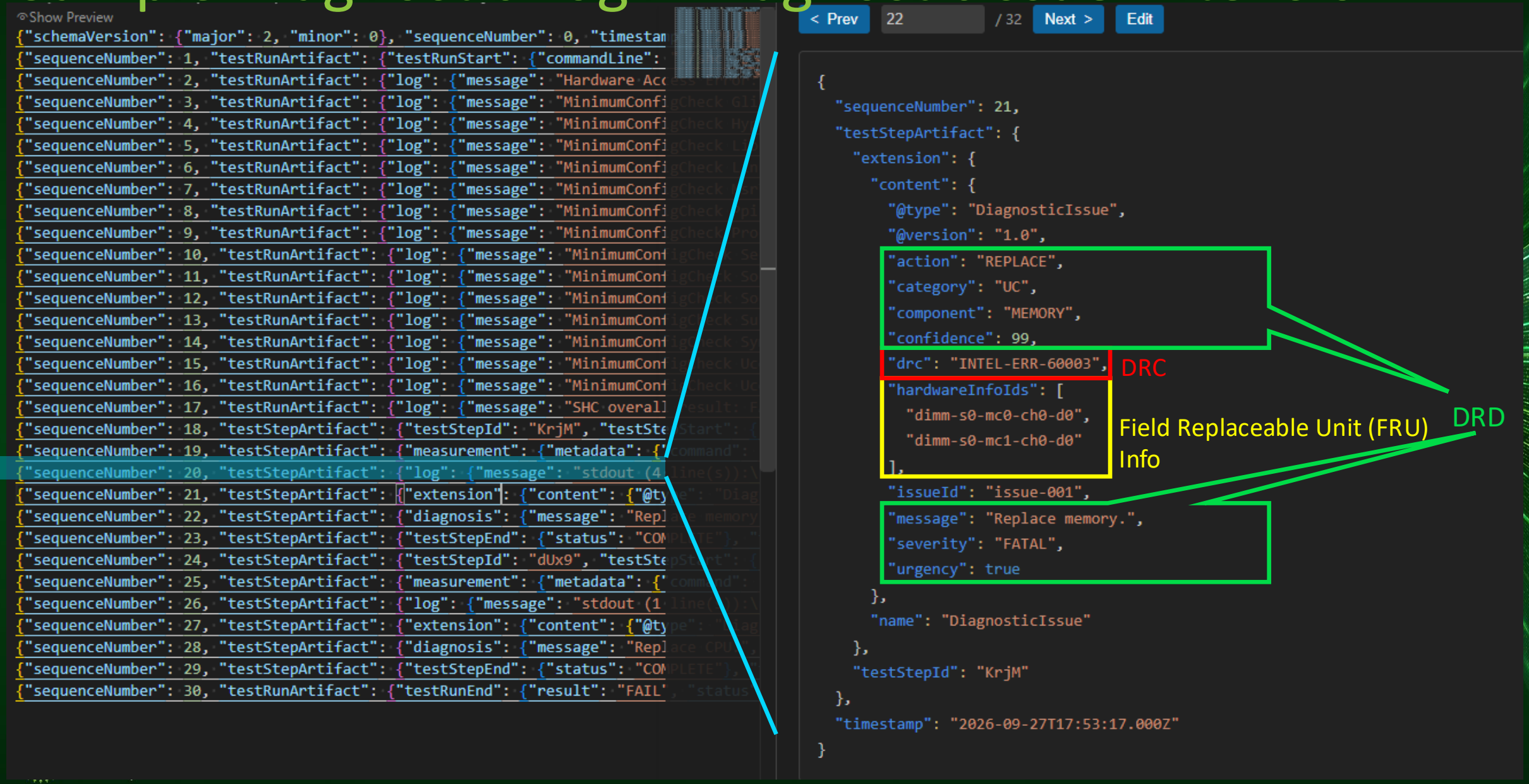
Sample DRC-DRD Map File

```
{
  "schema_version": "0.9",
  "map_revision": "2026.09.01",
  "DRCs": {
    "INTEL-ERR-10001": {
      "message": "Replace CPU.",
      "component": "PROCESSOR",
      "severity": "FATAL",
      "category": "UC",
      "action": "REPLACE",
      "urgency": true,
      "confidence": 99
    },
    "INTEL-ERR-20001": {
      "message": "Replace CPU.",
      "component": "PROCESSOR",
      "severity": "FATAL",
      "category": "SDE",
      "action": "REPLACE",
      "urgency": true,
      "confidence": 99
    },
    "INTEL-ERR-60003": {
      "message": "Replace memory.",
      "component": "MEMORY",
      "severity": "FATAL",
      "category": "UC",
      "action": "REPLACE",
      "urgency": true,
      "confidence": 99
    }
  }
}
```

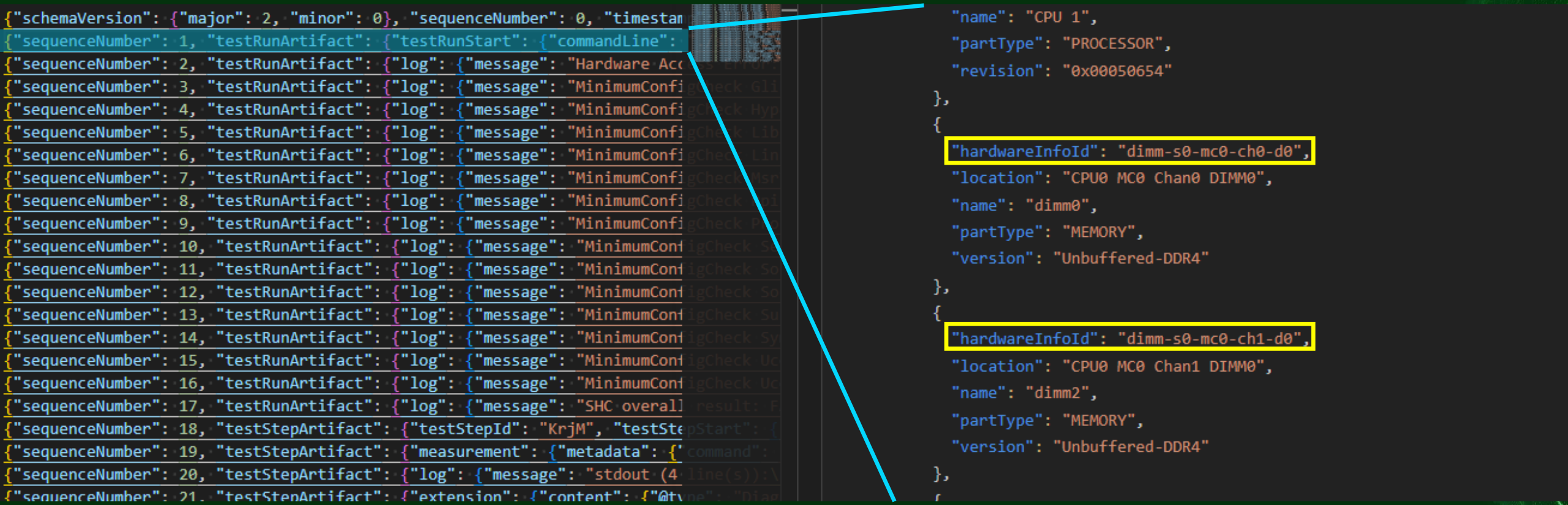
DRC

DRD

Sample Diagnostic Log – DiagnosticIssue Extension

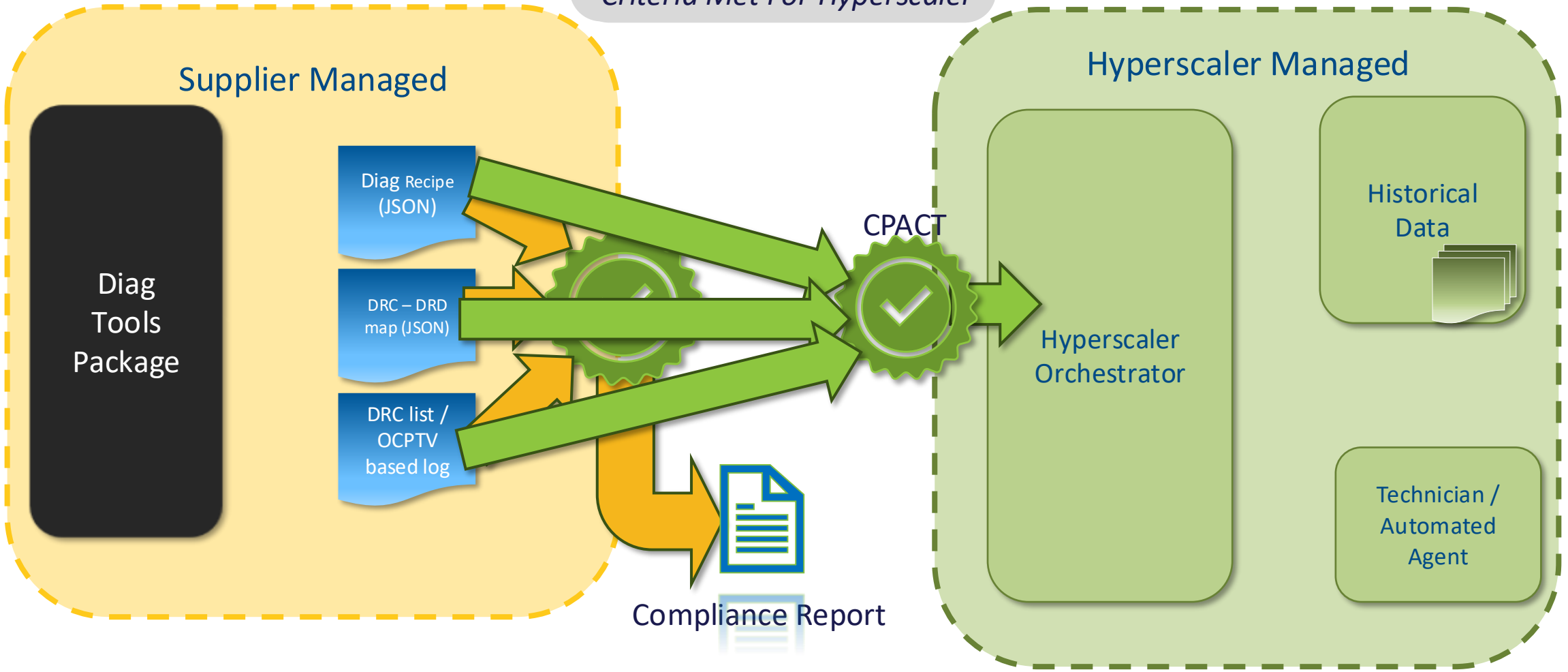


Sample Diagnostic Log – hardwareInfoId in dut_info



CPACT Role

Ensures Artifacts Check-In Criteria Met For Hyperscaler



Key Learnings

- **Standardized Diagnostics:** Standardize diagnostic artifacts enable consistent integration and evaluation.
- **Cross-Hyperscaler Alignment:** Shared specifications reduce provider-specific enablement effort.
- **Tiered Diagnostic Model:** v0.9 lays the groundwork for advanced telemetry-driven diagnostic workflows through a tiered diagnostic framework. By enabling progressive, condition-based fault isolation, the model supports faster root-cause identification and meaningful reductions in Mean-Time-To-Repair (MTTR).

Call to Action

- Problem to solve
  - Build scenarios and recipes for Hyperscale CPU Debug, Diagnostics, RAS, and Impactless FW Updates using CPACT
- How to get involved in the project/sub-project community
  - Contribute by commenting on our GitHub: opencomputeproject/ocp-diag-cpact
  - Reach out to OCPTV leads for any feedback
- Check out our specifications to learn more
  - https://tinyurl.com/OCP-Hyperscale-CPU-Debug
  - https://tinyurl.com/OCP-Hyperscale-CPU-Impactless
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Thank You!

